
VENTURE CAPITALIST GOVERNANCE AND VALUE ADDED IN FOUR COUNTRIES

HARRY J. SAPIENZA
University of South Carolina

SOPHIE MANIGART
University of Ghent, Belgium

WIM VERMEIR
University of Ghent, Belgium

EXECUTIVE SUMMARY

The rapid internationalization of markets for venture capital is expanding the funding alternatives available to entrepreneurs. For venture capital firms, this trend spells intensified competition in markets already at or past saturation. At issue for both entrepreneurs and venture capital firms is how and when venture capitalists (VCs) can provide meaningful oversight and add value to their portfolio companies beyond the provision of capital. An important way VCs add value beyond the money they provide is through their close relationships with the managers of their portfolio companies. Whereas some VCs take a very hands-off approach to oversight, others become deeply involved in the development of their portfolio companies.

Utilizing surveys of VCs in the United States and the three largest markets in Europe (the United Kingdom, the Netherlands, and France), we examined the determinants of interaction between VCs and CEOs, the roles VCs assume, and VCs' perceptions of how much value they add through these roles. We examined the strategic, interpersonal, and networking roles through which VCs are involved in their portfolio companies, and we analyzed how successful such efforts were. By so doing we were able to shed light on how and when VCs in four major markets expend their greatest effort to provide oversight and value-added assistance to their investment companies.

Consistent with prior empirical work, we found that VCs saw strategic involvement as their most important role, i.e., providing financial and business advice and functioning as a sounding board. They rated their interpersonal roles (as mentor and confidant to CEOs) as next in value.

Address correspondence to Harry J. Sapienza, College of Business Administration, University of South Carolina, Columbia, SC 29208.

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Finally, they rated their networking roles (i.e., as contacts to other firms and professionals) as third most important. These ratings were consistent across all four markets. VCs in the United States and the United Kingdom were the most involved in their ventures, and they added the most value. VCs in France were the least involved and added the least value; VCs in France appeared to be least like others in terms of what factors drove their efforts. Our theoretical models explained a greater proportion of variance in governance and value added in the United States than elsewhere. Clear patterns of behavior emerged that reflect the manner in which different markets operate. Among the European markets, practices in the United Kingdom appear to be most like that in the United States.

Determinants of Governance (Face-to-Face Interaction)

We operationalized VC governance or monitoring of ventures as the amount of face-to-face interaction VCs had with venture CEOs. We found some evidence that VCs increase monitoring in response to agency risks, but the results were mixed. Lack of experience on the part of CEOs did not prompt significant additional monitoring as had been predicted. A more potent determinant was how long the VC-CEO pairs worked together; longer relationships mitigated agency concerns and reduced monitoring. Contrary to expectations, perceived business risk in the form of VCs' satisfaction with recent venture performance had little impact on face-to-face interaction. Monitoring was greatest in early stage ventures, indicating that VCs respond to high uncertainty by increased information exchange with CEOs. We measured two types of VC experience and found different patterns for the two. Generally speaking, VCs with greater experience in the venture capital industry required less interaction with CEOs, whereas VCs with greater experience in the portfolio company's industry interacted more frequently with CEOs than did VCs without such experience.

Determinants of Value Added

We argued that VCs would most add value to ventures when the venture lacked resources or faced perceived business risks, when the task environment was highly uncertain, and when VCs had great investing and operating experience. Contrary to expectations, VCs added most value to those ventures already performing well. As we had predicted, VCs did add relatively more value when uncertainty was high: e.g., for ventures in the earliest stages and for ventures pursuing innovation strategies. Finally, we found that VCs with operating experience in the venture's focal industry added significantly more value than those with less industry-specific experience. These results are consistent with anecdotal evidence that entrepreneurs have a strong preference for VCs with similar backgrounds as their own. We found no evidence that experience in the venture capital industry contributed significantly to value added. Together, these results suggest that investigations of the social as well as economic dimensions of venture building may prove a fruitful avenue for future study. Overall, the results showed that value-added is strongly related to the amount of face-to-face interaction between VC-CEO pairs and to the number of hours VCs put in on each individual venture.

Implications for Venture Capitalists

The competition for attractive investments is heating up as economies become more globalized. Thus, the pressure on venture capital firms to operate both efficiently and effectively is also likely to build. It is as yet unclear whether the recent trend toward later stage, safer investments will continue, and how those venture capital firms following this path can differentiate themselves from other sources of capital. Venture capital firms that are able to choose the appropriate bases for determining governance effort and the appropriate roles for delivering added value to their portfolio companies will be those most likely to survive.

In the largest, most robust markets (i.e., the United States and the United Kingdom), more effort is expended by venture capitalists to deliver something of value beyond the money. This sug-

gests that the tradeoff preferred by those succeeding is to be more rather than less involved in their investments. Our results indicate that VCs clearly economize on the time they devote to involvement in their portfolio companies. However, our results also indicate that they do this at the great peril of producing value insufficient to justify the cost of their product.

Implications for Entrepreneurs

Our findings provide two important insights for entrepreneurs. First, they show that where and when they obtain venture capital is likely to have an impact on the extent and nature of effort delivered by their venture capital investors. It appears that on average entrepreneurs receiving venture capital in the United States and the United Kingdom will be more closely monitored and will receive more value-adding effort from their VCs than will those in France or the Netherlands. Needless to say, entrepreneurs should consider their preferences for level and type of involvement from their investors as they consider their choice of partners. In France, for example, VCs put great emphasis on their financial role in comparison with other roles, but they contribute much less than VCs elsewhere via other strategic, interpersonal, and networking roles.

The second key implication of our findings is that entrepreneurs may be able to gauge what roles VCs will see as most important, when VCs are more or less apt to become involved in their companies, and when they believe they can most add value. Such knowledge may help CEOs anticipate VC activity, be aware of the parameters of VCs' preferences, communicate their own preferences, and negotiate the timing and extent of interaction. For example, although our results indicate that geographic distance significantly limits face-to-face interaction, it appears to have less impact on the amount of value added.

Implications for Researchers

Much more can be learned about the relative efficiency and effectiveness of alternative governance arrangements. Little is known about how formal structures such as contract covenants and board control work in conjunction with informal oversight and interaction. Even less is known about how value is added and how it is best measured. Although this study took a step toward developing a model of the circumstances under which value is added, the theory and its operationalization await further development. ©1996 Elsevier Science Inc.

INTRODUCTION

Although venture capital developed in the United States decades before it did elsewhere, the last 15 years have witnessed explosive growth in other venture capital markets, especially those in Europe (Bygrave and Timmons 1992; Ooghe, Manigart, and Fassin 1991). Some researchers have speculated that venture capital markets in Europe have evolved in a pattern similar to that in the United States but have also noted structural differences between U.S. and European venture capital even from the outset (Ooghe et al. 1991; Sapienza, Amason, and Manigart 1994). Of considerable practical importance to venture capitalists (VCs), entrepreneurs, and policy-makers is whether the differences in these markets translate into systematic differences in VCs' efforts to govern their portfolio companies or in their efforts to add value beyond the money they provide. This study therefore was designed to examine the drivers of venture capitalist governance and value added in the United States and in the three largest markets for venture capital in Europe (the United Kingdom, France, and the Netherlands).

This study should hold practical value for several reasons. First, knowledge of VCs' practices are of interest to entrepreneurs attempting to select the appropriate source of fund-

ing. The choice will involve both the type of funding to seek and where and when to seek venture capital funding. Advancements in communication technology and the internationalization of markets appear to make cross-border choices more feasible than ever. For venture capital firms (VCFs), knowledge of others' practices could prove useful in selecting their domain, planning their portfolio, choosing which competencies to develop, and setting operating policies. Finally, policy-makers could learn from the practices used in various markets how they might enhance VCFs' efforts to fuel the engines of economic growth and success.

In a recent article, Amar Bhidé (1994, p. 139) claimed that "entrepreneurial, privately held companies have taken over as the engines of innovation and job growth." Bhidé's view is that the ascension of the entrepreneurial firm in the United States is at least partially the result of public policies that prevent the development of close relationships between outside investors in large, public firms and the firms' inside management. In contrast, private, entrepreneurial ventures may establish personal relationships with investors such as VCs who "try to increase companies' value through counseling and oversight" (p. 138). Indeed, empirical evidence shows that VCs are much more involved in their investments than outside owners in public corporations (Rosenstein 1988). However, evidence also shows that VCs' involvement varies from near passivity to deep involvement (Gorman and Sahlman 1989; MacMillan, Kulow, and Khoylian 1989), and some controversy still exists as to how or even whether VC involvement adds value to new ventures (Rosenstein et al. 1993; Sapienza 1992).

Studies of VC activity show that VCs are often actively and personally involved in their portfolio companies (Bygrave and Timmons 1992; Fried and Hisrich 1995; Rosenstein 1988). Their involvement includes activities to monitor and provide assistance¹ to venture management (Gorman and Sahlman 1989; MacMillan et al. 1989). However, whereas such personal contact may afford the opportunity to reduce risks or add value, greater governance may not always be cost effective. Barney et al. (1989, p. 64) point out that VCFs must balance the costs of governance against the benefits. Similarly, Gorman and Sahlman (1989, p. 241) observe that VCs seek to maximize the return on the time they put into ventures. These studies recognize the tradeoffs that VCs make, but neither one explicitly examines how they determine the appropriate level of personal involvement. MacMillan et al. (1989, p. 37) observe that "a relevant issue in need of examination is the opportunity cost of [greater] involvement."

Because the value of investors' personal involvement in investments may vary with circumstances, we designed this study to examine the conditions under which VCs become more involved in their portfolio companies and those under which they believe their involvement is of greatest value to these companies. Recent works in the field have applied economic and organizational theories to model variations in VCF governance (Barney et al. 1989; Sapienza and Gupta 1994). We argue here that to understand VCs' choices regarding the amount of effort to expend interacting with CEOs it is important to consider the impact of the delegation of authority to the CEO (an agency issue), expectations regarding returns on investment (a business risk issue), uncertainty regarding the decision-making capacity of the VC-CEO pair (an information-processing issue), and the experience level of the VCs

¹We define VC governance as the oversight (Lerner 1995) and monitoring of portfolio companies. We follow Alchian and Demsetz's (1986, p. 135) broad definition of what monitoring might entail: "We use the term monitor to connote several activities in addition to its disciplinary connotation. It connotes measuring output performance, apportioning rewards, observing the input behavior of inputs as a means of detecting or estimating their marginal productivity, and giving assignments or instructions in what to do and how to do it. . . ."

themselves. We also examine how and when VCs add value to their portfolio companies. Because no theory of venture capitalist value added exists, we developed a framework that posits that value added is greatest when ventures are struggling, when uncertainty is great, and when VCs have experience pertinent to the ventures.

This research represents the broadest comparative study of this nature to date. The use of four markets provides descriptive information of practical value to entrepreneurs, VCFs, and policy-makers and provides insights into the generalizability of the theories forming the framework of the study. A portion of the data we utilize have been reported in prior studies (Sapienza 1992; Sapienza et al. 1994; Sapienza and Timmons 1989), but this study also provides new data and analyses. Differences and overlap between this and past studies are elaborated in the Method section.

THEORETICAL FRAMEWORK

Since Berle and Means's (1932) seminal work, considerable research in both organizational and financial economics has addressed the question of the ability of a firm's owners to monitor and influence the actions of its senior executives (Eisenhardt 1989; Oviatt 1988; Walsh and Seward 1990). Although most of these studies have been conducted in the context of large publicly held firms, Eisenhardt (1989) has suggested that these theories could be productively applied also to the governance of smaller, newer firms where there is less dispersion of ownership, where the inherent level of outcome and uncertainty is greater, and where there is a weak market for external control. Thus, closely held firms in developmental stages, especially those that have not yet entered the public market, offer a potentially fruitful arena in which to examine agency and organization theory perspectives on governance and decision-making.

Research on Venture Capitalist Involvement in Portfolio Companies

Studies of VC activity and value added have tended to be descriptive and somewhat atheoretical. Two key studies are those of Gorman and Sahlman (1989) and MacMillan et al. (1989). Gorman and Sahlman found that senior partners in VCFs spend 60% or more of their time in postinvestment activities and contribute to ventures via financing, strategic and operational planning, management recruiting, and providing a network of contacts. MacMillan et al. (1989) built on Gorman and Sahlman by examining a more elaborate list of 20 activities. Both studies found that VCs rate their strategic activities as the most important. In the MacMillan study, the role of "sounding board" was rated highest; the authors speculated that this might include acting as confidant or mentor, but following Gorman and Sahlman, they did not ask VCs to rate such roles. Gorman and Sahlman found VCs most active in early stage ventures, but MacMillan et al. observed no correlation between activity and context and concluded that VC involvement is primarily a matter of personal preference or inclination.

VCs in the MacMillan et al. study were generally quite satisfied with their involvement in their portfolio companies. However, whether or not VCs actually add value through involvement remains controversial (Sapienza 1992). For example, MacMillan et al. observed both positive and negative associations between involvement and venture performance; Rosenstein et al. (1993) found that entrepreneurial CEOs rate VCs no more valuable than other outside investors, unless one of the "top 20" VCFs is represented on the board of directors.

The few studies of European VC activity parallel the U.S. studies but also provide in-

sight into the social roles VCs play. Using the MacMillan et al. (1989) framework, Harrison and Mason (1992) found that entrepreneurs in the United Kingdom rate sounding board and strategic activities as most important and that they depend more on the assistance of VCs than they do on other types of investors. A British Venture Capital Association survey (1992) found that prior to investment, entrepreneurs express a preference for passive financial assistance only, but postinvestment, they rate active, nonfinancial assistance as providing the highest value added. In the Netherlands, VanWakeren, Van der Werf, Vonk, and Schuit (1989) found that about 60% of entrepreneurs receive important strategic assistance from VCs, and about 20% receive useful networking assistance. Studies in Nordic markets reveal the following: (1) in Finland, the activities of financing and active oversight are rated most important (Ahdekivi 1990); (2) in Sweden, one survey found entrepreneurs rate access to capital, to VC competence, to moral support, and to networks most important (Fredriksen et al. 1990); and (3) another survey in Sweden observed four styles of venture VC involvement: consultancy oriented, mentor oriented, operations oriented, and structure oriented (Landstrom 1991). The two studies in Sweden give explicit recognition to an issue raised, but not explored, in the early U.S. studies: the importance of the personal relationship between the VC and the entrepreneur.

The importance of the relationship between the VC and the entrepreneurial CEO has also begun to receive attention in the United States. For example, Barney et al. (1994a) observed that unless the entrepreneur is receptive to VC advice, no value added can be achieved through VC involvement, and Fried and Hisrich (1995) went so far as to label the VC a "relationship investor."

Determinants of Governance through Interaction

This new attention to VC-entrepreneur relations provides another way of viewing VC governance. The work of Barney and colleagues (1989, 1994b) shows that VCFs systematically adjust structural governance mechanisms to match agency and business risk. The first of their studies showed that the more the CEO and the managerial team have at stake in the venture and the better the venture's performance in the year of first-round financing, the less VCFs invest in structural governance mechanisms such as presence on the venture's board of directors. The second study found that if new firms experience high returns or if there are obstacles to monitoring, it is more likely that contracts will include covenants that limit venture management's discretion. But structural and contractual devices are crude mechanisms of control (Walsh and Seward 1990). A premise of this study is that VCs influence entrepreneurial behavior through their personal contact with entrepreneurial CEOs. However, because personal contact involves significant costs, we expect that VCs will consciously choose how much effort to expend depending upon their experience and their perceptions of the agency risk, business risk, and uncertainty inherent in the situation.

Personal interactions with CEOs may serve several purposes for the VC: they allow the VC to become personally familiar with what is going on in the business, to understand how the CEO operates and thinks, to build rapport, and to influence decisions. Nonetheless, because time is a precious resource for VCs (Gorman and Sahlman 1989) and because entrepreneurs may resent excessive intrusion (Sapienza 1992), VCs will seek to limit face-to-face contact to the extent possible (Barney et al. 1989). Thus, efficiency is an underlying principle in our governance hypotheses. Given that general partners are involved on average in six or more portfolio companies (Gorman and Sahlman 1989), efficiency of oversight appears a critical issue for VCFs (Lerner 1995).

Impact of Agency Risks on VC-CEO Interaction

We expect that VCs will vary their personal contact with CEOs depending on the agency risk inherent in the context. We define agency risk as the risk that CEOs (agents) will take actions contrary to the wishes of their VCs (Jensen and Meckling 1976). Such risks would be higher either when CEOs have greater incentive to pursue self-serving behavior at the expense of venture outcomes (Barney et al. 1989) or when VCs lack complete faith in the judgments of the CEOs (Sapienza and Gupta 1994). Because CEOs generally hold significant equity stakes in their ventures and have great incentive to maximize venture outcomes (Sapienza and Gupta, 1994), we focus here on agency risks emanating from VC uncertainty regarding CEO judgment rather than on those associated with managerial opportunism.

As noted by Sapienza and Gupta (1994), although VCFs expend great effort to ensure the quality of venture management before investment, the variety of demands on the CEO make it virtually impossible that any CEO or management team will be “perfectly” competent in all situations. Some CEOs may have great technical experience, whereas others understand the venture building process thoroughly. VCs’ continuing doubts are reflected in both Ruhnka and Young’s (1987) finding that managerial competence is among the greatest of VCs’ concerns and Gorman and Sahlman’s finding that VCs overwhelmingly attribute venture failure to managerial incompetence; indeed, Gorman and Sahlman speculate that the nature of the venture capital process itself requires that VCs be constantly vigilant with regard to potential managerial weaknesses. Thus, a critical paradox arises: VCs must both support and question management decisions.

In short, it appears likely that VCs will expend additional effort to interact on a face-to-face basis when they fear that a CEO’s competence poses a threat to the venture’s success. This reasoning is consistent with Fiet’s (1995) view that, postinvestment, VCs manage agency risks through face-to-face interaction, a form of contact that provides an opportunity to gather information and to create a personal understanding of the business and its people. However, the greater VCs’ confidence in CEOs’ decision-making, the less their incentive to incur the costs of close, personal governance. Thus, we offer our first hypothesis:

H1: The higher the agency risks, the greater will be the face-to-face interaction between the VC and the CEO.

Impact of Perceived Business Risk on VC-CEO Interaction

Barney et al. (1989, p. 64) define business risk for VCFs as “the uncertainty associated with obtaining returns on an investment in a new firm due to that firm’s competitive environment.” They show that as such uncertainties increase, VCFs increase the elaborateness of their governance mechanisms to protect against adverse outcomes. Consistent with their work, we expect that VCs will also increase face-to-face governance of the CEO when business risk is high.

Because the extent of business risk cannot be perfectly anticipated in advance, VCs will adjust their level of monitoring to fit the perceived progress of the ventures. We posit that when ventures are exceeding expectations, VCs will be less inclined to be involved. As an example, one of the VCs we interviewed said, “When the venture is really moving, the best thing you can do is to just stay out of the way of the freight train.” On the other hand, it seems likely that when VCs believe the ventures are doing very poorly they have a fiduciary responsibility to find out why. Whereas VCs’ incentive to expend the time and effort required for face-to-face interaction with CEOs is low for ventures meeting or exceeding

targets, VCs of poorly performing ventures are likely to become more involved, if only to determine whether or not venture management should be changed. Thus, we offer our second hypothesis:

H2: The higher the perceived business risk faced by the venture, the greater will be the face-to-face interaction between the venture capitalist and the venture CEO.

Impact of Uncertainty on VC-CEO Interaction

VC oversight of ventures involves not only the monitoring of CEOs' decisions but also the direct involvement of VCs in making strategic decisions (Gorman and Sahlman 1989; Lerner 1995; Rosenstein 1988). Sapienza and Gupta (1994, p. 1622) argued that the information-processing perspective (Galbraith 1973) could provide useful insights as to why VCs might seek more or less interaction in assisting CEOs in decision-making. The value that this perspective adds is that it provides insights as to why VCs might vary monitoring efforts even when there are no conflicts of interest among decision-makers. The information-processing theory holds that the structure of joint decision-making depends upon the level of uncertainty (i.e., the gap between the information necessary to make effective decisions and the information already possessed by the decision-makers [Galbraith 1973]). Specifically, it suggests that decision-makers will seek cost-efficient means to reduce uncertainty.

From an information-processing perspective, two basic alternatives exist for reducing uncertainty: decision-makers can reduce goals or they can increase their information-processing capacity. VCFs do not invest money in high risk ventures with the idea of obtaining modest but certain returns. Because VCFs seek high returns on their capital (Bygrave and Timmons 1992), it is the second of these two rationales that is apropos to most VC-CEO pairs. Thus, we expect that as the uncertainty facing ventures increases, VCs will attempt to increase ventures' information-processing capacity by stepping up their face-to-face interaction with the CEO. This expectation can be stated formally as follows:

H3: The higher the uncertainty facing the venture, the greater will be the face-to-face interaction between the venture capitalist and the venture CEO.

Impact of VC experience on VC-CEO interaction.

Directly bearing on the issue of the amount of uncertainty facing the venture is the information-processing capacity of the VC-CEO pair responsible for many of the key decisions facing the venture. All other things being equal, we expect that the amount of information VCs bring to the table will increase directly with their experience as venture capitalists or in the venture's focal industry. Further, we also expect that greater experience will be associated with greater efficiency in information processing and decision-making. Thus, for highly experienced VCs, less face-to-face interaction should be required to produce decisions of the same quality as those produced by less experienced VCs (Sapienza 1989; Williamson 1975). Given that VCs will seek to economize on the time consumed in the direct governance of ventures (Barney et al. 1989; Fiet 1995), these arguments suggest that experienced VCs are more likely to be able to economize and still adequately perform their fiduciary responsibilities. These expectations can be formally stated as the following hypothesis:

H4: The greater VCs' experience, the less frequent will be the interaction between the venture capitalist and the venture CEO.

Determinants of Value Added

Theories of how and when VCs most add value to the ventures they fund are not well-developed. We take steps toward a theory of investor value added by developing expectations and testing them on our data set. We will rely on the information-processing perspective (Galbraith 1973) as the basis for making predictions regarding when VCs most add value to the ventures in which they invest. Our central thesis is that when CEOs lack sufficient information or information-processing capacity or when they perceive great information potential in their VCs they will seek the assistance of their VCs. Three broad conditions are expected to increase VCs' value added: (1) when venture capacities fall short of venture "needs," (2) when the venture faces great uncertainties regarding what the optimal strategies are or how best to implement the chosen strategies, and (3) when the VCs themselves appear to offer a significant source of valuable information for the venture. These views are also consistent with resource dependency theory (Pfeffer and Salancik 1978): VCs will be most valued when they are able to reduce critical uncertainties for the venture (Bygrave and Timmons 1992). This greater contribution will result not only because VCs may help substantively in the reduction of uncertainties, but also because the CEO is likely to ask for more help in these situations and to be more receptive to the advice given by the VC (Barney et al. 1994a).

Impact of Venture Needs on Value Added

Lerner (1995, p. 301) argued recently that VCs provide the greatest assistance to new ventures "when the need for oversight is great[est]." On the surface, this view may appear a near tautology. Surely, it makes great sense to argue that a venture can most benefit from outside advice when it is "neediest." Won't the venture benefit most from VC involvement when the venture management team is incomplete or inexperienced or when the venture is struggling to survive? Deeper consideration of the issues, however, suggests that the converse is possible. Sapienza et al. (1994) speculated that VCs might follow "home-run" strategies, preferring to put greater effort and resources into ventures promising the greatest payoff. Similarly, experienced venture CEOs and management teams less in need of outside assistance may be more capable of evaluating and leveraging the input of their VCs.

If we view the CEO-VC pair as a key decision-making unit for the venture, the information-processing perspective implies that as the venture's information-processing capacity falls below that required for effective decisions, VCs will be motivated to increase their input. Inexperienced CEOs are likely to have fewer contacts to critical information and to be subject to greater equivocality regarding cause and effect relations than are more experienced CEOs. The duration of the VC-CEO relationship should also have had an impact on the amount of value added during any given interval of time: the less time the venture had been in the portfolio, the less likely that the venture will have absorbed all knowledge, reputation, and network benefits offered by VCF-backing. Thus, the greatest value added probably occurs earlier on in the relationship. Finally, ventures seen to be performing poorly may attract the greatest concern of investors fearful of realizing a total loss on investment. An additional reason to expect that VCs' value will be greatest under these circumstances is that entrepreneurs may be more receptive to VCs' input when they perceive a greater need for assistance. In summary, we expect value added to be high when CEOs are less experienced, when the relationship is older, and when the venture is seen as performing poorly. These arguments lead to the following hypothesis:

H5: The greater the needs of the venture, the greater will be the value added of the VC's involvement.

Impact of Uncertainty on Value Added

Sapienza and Gupta (1994) found that the greater the uncertainties facing the venture, the greater was their face-to-face interaction. They attributed this finding both to the increased difficulty of assessing outcome causes in highly uncertain situations and to the desire on the part of both parties to improve decision-making. This interpretation is consistent with the view that a viable option for decision-makers facing great uncertainty is to increase lateral relations (Galbraith 1973). An assumption of this perspective is that decision-makers are limited in their rational capacities. The extent of limitations on individual decision-makers is a direct function of the complexity and novelty of the problems they face. When CEOs face very simple, stable conditions, the cost of interacting with outside advisors may exceed the benefits. However, we expect that as problem complexity and novelty rise, so too will the value of interacting with outside advisors. Thus, we offer the following hypothesis:

H6: The greater the uncertainty facing the venture, the greater will be the value added of the VC's involvement.

Impact of VC Experience on Value Added

VCs' experience may be the most critical factor determining the amount of value added received through interaction. As was the case with venture need, the expected impact of VCs' experience on value added may appear obvious. Clearly, VCs who have financed and monitored hundreds of new ventures, have started some of their own, or who have years of experience in the ventures' industries are likely to have more to offer CEOs than those who have limited experience. Further, is it not likely that CEOs will be more receptive to the views of VCs with great experience? Indeed, we expect VC experience to be positively related to value added, but we also believe some caveats are in order.

Implicit in most of our arguments has been the view that greater effort on the part of VCs is associated with greater value added. However, earlier we argued that experienced VCs could satisfy their fiduciary responsibilities with less effort. Thus, the possibility exists that experienced VCs will interact less with CEOs and will therefore add less value to the ventures than inexperienced VCs. Anecdotal evidence suggests that if VCs' greater experience is as a financier of new ventures (i.e., as a venture capitalist) rather than as a practitioner in the venture's industry, the danger of resentment on the part of the CEO is great (Sapienza 1989). Indeed, we heard complaints in interviews with both VCs and entrepreneurs about VCs who were "financial MBA-types" or who "never started any business of their own."

Although the above dangers exist, we expect that VCs who survive to become experienced are those who are aware of these perceptions and are able to manage involvement so as to contribute value beyond the money provided. Rosenstein et al. (1993) found that VCs from the top U.S. VCFs were perceived as adding more value than other outsiders. The importance of reputation and experience are implicit in their arguments and supported by their findings. In summary, we expect that VCs with experience in the venture's industry will have a great deal to offer because of their understanding of the business, whereas VCs with a great deal of experience as venture capitalists will have much to offer because of their understanding of the venture development process. Further, though this latter group may be especially inclined to try to economize on effort, they will not do so at the expense of offering more than enough to justify the price entrepreneurs pay for their investment. Thus, our final hypothesis is as follows:

TABLE 1 Venture Capital Markets in Four Countries^a

	U.K.	FR	NL	U.S.
Funds raised				
(absolute):	1249	857	91	\$2.5B
(as a percentage of the country's gross domestic product):	0.15%	0.08%	0.04%	0.04%
Funds invested				
(absolute):	1831	968	239	\$2.5B
(as a percentage of the country's gross domestic product):	0.23%	0.09%	0.10%	0.04%
Percentage of funds invested by VCF type				
Independent investors	55	33	61	81 (private)
Captive investors	33	38	27	18 (corporate)
Semicaptive investors	11	28	0	1 (SBICs)
Public	1	1	12	

^a All figures for Europe are for 1992 and come from EVCA (1993); Funds Raised and Funds Invested in Europe are in millions of ECUs. For the U.S., Funds Raised and Funds Invested are for 1992 in billions of \$ (source, *Pratt's Guide to Venture Capital Sources* (1995). Percentage of Funds Invested by VCF Type in the U.S. represent the period 1986–1989 and Percentage of Funds Invested by Stage is for 1989 [source: Bygrave and Timmons, 1992]. The labels in parentheses reflect the terminology in the U.S. when it differs from that in Europe.

U.K. = United Kingdom; FR = France; NL = the Netherlands; U.S. = United States.

H7: The greater the experience of the VC, the greater will be the value added of the venture capitalist's involvement.

INDUSTRY CONDITIONS IN THE FOUR COUNTRIES

Industry structure and investment patterns within countries may contribute to differences in how involved VCs are in their portfolio companies. Although Europe is one of the fastest growing venture capital markets in the world (Bygrave and Timmons 1992), relatively little empirical work has examined the European venture capital industry (Ooghe et al. 1991). Ooghe et al. observe that not only are the venture capital industries in the United States and Europe structurally different from one another, but that significant differences also exist among markets within Europe. For example, VCFs' sources of funds, their preferences for stage and industry sector, the fiscal climate, and even the extent of VCF market dominance may vary on a country-by-country basis. Therefore, we will briefly describe the venture capital markets in the four countries at time the data collection was done (early 1988 in the United States and mid-1992 in Europe).

Readers should keep in mind not only the difference in the time period of data collection but also differences in terminology across settings. Bygrave and Timmons (1992, p. 73) caution that definitions of venture capital vary even within countries, much less on a worldwide basis; for example, they point to the fact that what 3i (the dominant VCF in the UK) would call a "start-up" would not be considered one in the United States. Table 1 presents statistics on capital raised and invested in each market and the percentage of funds invested by VCF type.

Whereas the venture capital industry in the United States was larger and more developed than that of any other individual country, figures show that by 1988 Europe had caught up with and surpassed the United States in terms of funds raised and invested annually (Bygrave and Timmons 1992; Ooghe et al. 1991). The European venture capital industry reached its peak in 1989 in terms of total new funds raised for venture capital (5.8 billion ECU): this

stabilized at 4.2 billion ECU in 1991 and 1992. The yearly amount invested, however, grew slowly in Europe: 4.3 billion ECU in 1989 to 4.7 billion ECU in 1992 (EVCA 1993, p. 63). The European VC industry clearly came to maturity in 1992: divestments in 1992, measured at cost, totalled 2.3 billion ECU or 17% higher than in 1991. One major difference from the U.S. venture capital industry is that trade sales in Europe remain the most important means of divestment: they account for 41% of 1992 divestments, whereas public offerings accounted for only 17% of divestments in 1992 and under 10% in 1991 (EVCA 1993, pp. 47–48). Europe lacks active IPO markets as exit routes for venture capital investments (Leloux and Manigart 1994).

In absolute numbers, the United Kingdom, France, and the Netherlands were the largest venture capital markets in Europe. Table 1 shows that the United Kingdom held a healthy lead over France, which in turn was a much larger market than the Netherlands. The venture capital industry in Europe comprised four distinct groups of organizations (EVCA 1993): (1) independent firms; (2) captive organizations, part of larger financial institutions; (3) semicaptive organizations, partially owned by larger (financial) institutions but operating independently; and (4) public firms.² In the United States, the most common way of classifying VCFs was as private firms, SBICs, and corporate subsidiaries (*Pratt's Guide to Venture Capital Sources* 1988). In 1992, almost two-thirds of the investments were made by independent organizations in the Netherlands, compared with 54.7% in the United Kingdom, and only 32.3% in France. Whereas the U.S. market in 1988 was also dominated by large, private VCFs (81% of capital under management in 1989 was held by private VCFs, and 14% of all VCFs controlled 59% of the funds [Timmons and Sapienza 1992]), no one firm in the United States held the dominating presence that 3i did in the United Kingdom (Ooghe et al. 1991).

In 1992, the amount of funds raised by VCFs had decreased slightly in the Netherlands and in France, but had remained relatively constant in the United Kingdom (EVCA 1993). In the United States, the \$4.2 billion raised in 1987 was a high point. However, Black Monday in October of that year marked a turning point for venture capital in the United States; only \$2.9 billion was raised in 1988, beginning a downward spiral that reached its nadir in 1991 at \$1.3 billion before rebounding to \$3.8 billion in 1994 (*Pratt's Guide to Venture Capital Sources* 1995). Table 1 indicates the amount of venture capital raised and invested in 1992 in the four markets as a percentage of each country's gross domestic product; these comparisons provide one view of the relative importance of venture capital in each country's economy. As may be observed, venture capital raised and invested in the United Kingdom (0.15% and 0.23%, respectively) represents a much larger proportion of that country's economy than does venture capital elsewhere. Although representing the single largest market in absolute terms, the U.S. market for venture capital is the smallest relative to its overall economy.

The most important sources of funds for independent VCFs in the United Kingdom and in the Netherlands were the pension funds and insurance companies, whereas institutional investors were relatively absent from the venture capital market in France. Funds raised from institutional sources in France tended to come from abroad (EVCA 1993). Moreover, in France, financial institutions and banks dominated the industry by setting up captive and

²No formal definition of each distinct group exists. We use "independent firm" to refer to a firm that is not controlled by or set up by another financial or nonfinancial company; a "captive firm" is a firm that is completely controlled by and receives its capital for investments from a single parent company; a "semicaptive firm" is a firm that is completely controlled by the parent company who set up the firm, but in which other companies or private individuals provide part of the investment capital; finally, a "public firm" is set up by and controlled by the government. Therefore, Table 1, which is purely descriptive, explicitly states the difference in U.S. and European terminology where appropriate and the distinction by VC fund type is not further used in analyses.

TABLE 2 Venture Capital Firms and Portfolio Companies in This Study

	U.K.	FR	NL	U.S.	Total
VCF characteristics:					
Number of respondents	76	43	37	65	221
Number of affiliates	17	26	13	NA	NA
Number of government agencies	6	0	2	NA	NA
Venture board characteristics:					
Number of members of the board of directors	5.36	5.81	3.25	5.71	5.16
Inside members	3.26	2.64	1.03	1.96	2.43
Venture characteristics:					
Number of entrepreneurs with start-up exp.	26	17	12	25	80
Number of ventures by stage ^a					
Early stage ventures	20	6	4	29	59
Late stage ventures	55	37	43	36	161
Number of ventures by sector ^b					
Manufacturing	18	7	9	26	60
Energy/nat. resource	5	1	1	—	—
Medical/health	8	1	0	—	—
Agriculture	0	2	0	—	—
Consumer products	9	7	3	—	—
Computer related	15	7	1	15	38
Biotechnology	1	1	4	—	—
Chemicals/plastics	3	3	4	—	—
Financial services	2	0	0	6	8
Electronics	6	3	4	—	—
Percentage of a venture's competitors with lower levels of innovation ^c					
Technological	73	86	88	56	72
Marketing	59	63	57	43	53
Product	55	49	31	52	48

^a Differences between stage variables and innovation variables here and those in Table 4 reflect the fact that cases with missing values for any variable were dropped in the regression analyses.

^b Economic sectors were conceptualized differently in the U.S. studies so that comparable figures do not exist across sectors marked "—".

Abbreviations as in Table 1.

semicaptive venture capital organizations. In the Netherlands, the government directly invested in venture capital; this was not the case in the United Kingdom or in France. A new phenomenon in the United Kingdom was that funds were increasingly being raised for investing in foreign markets, especially targeted for southern Europe.

Pension funds were the dominant suppliers of capital in the United States during the period of the study (46%), whereas the remaining capital was raised from foreign investors (14%), endowments and foundations (12%), corporations (11%), insurance companies (9%), and individuals and families (8%); banks could not invest in venture capital in the United States (Timmons and Sapienza 1992).

There was much less investment in early stage and high tech businesses in European countries than in the United States (Murray 1991; Ooghe et al. 1991). Management buyouts and buyins accounted for almost two-thirds of the amount invested in the United Kingdom; in the Netherlands and France, expansion stage financing and replacement capital was prevalent. Among the European countries of this study, only in the Netherlands was as much as 8% of investment going to very early stage companies, whereas the figure was about 28% in the United States (Timmons and Sapienza 1992); again, however, differences in terminology pose a challenge to precise comparison.

Compared with all the countries in Europe, the United Kingdom and France have the

best environmental conditions (government incentives, tax structures, etc.) for venture capital, followed by the Netherlands (Tyebjee and Vickery 1988; Roure, Keeley, and Van der Heyden 1990). It is difficult to compare European fiscal policies directly with those in the United States, but some researchers have concluded that domestic VCFs in the United States are adversely impacted (in comparison with foreign entry) by fiscal policy (Timmons and Sapienza 1992).

Overall, the U.K. market appears most similar with respect to size, maturity, and dominant type of VCF to that of the United States. This is not surprising given that the U.K. market developed sooner than other European markets. However, vast differences between the two exist in terms of funding preferences and governance styles. Within Europe, the Netherlands and the United Kingdom look alike with respect to the industry structure, whereas the Netherlands and France look alike with respect to investment patterns. At the time of survey administration in the United States, the market for IPOs had just experienced a setback reflected in the severe drop in venture capital funds raised. In contrast, the European venture capital industry was relatively stable at the time the survey was administered, and European VCFs tended to invest in later stage, safer ventures (Bygrave and Timmons 1992).

METHOD

The data reported in this study were obtained in two separate collection efforts, the second a follow-up and replication of parts of the first (U.S.) study. The U.S. data were collected in 1987–1988; the procedures, sample, and methods involved in this study have been reported in detail elsewhere (Sapienza 1992; Sapienza and 1994; Sapienza and Timmons 1989). In brief, the U.S. study involved extensive interviewing of VCs and CEOs of their portfolio companies and follow-up questionnaires based on the interviews and prior empirical work. The personal interviews created a very high response rate (85% and 80%, respectively) and allowed us to develop our surveys; they also provided an in-depth view of the nature of the economic and personal relationships involved. Questionnaire responses were received from VCs on 65 of their portfolio companies; matched responses were obtained from CEOs of 51 of these ventures. High correspondence between VCs and CEOs regarding the importance and value of venture capitalist involvement in the ventures as well as venture performance allowed the responses to be combined as reliable, valid measures for 51 ventures; only these 51 were used in the earlier studies. The study reported here includes all 65 responses from VCs.

Because of the interrater reliability evident in the first study and because resources prevented obtaining matched responses of VC-CEO pairs in the three European countries, the follow-up study obtained survey responses from VCs only. However, this procedure allowed us to survey the entire target population. A slightly modified version of the U.S. questionnaire was prepared in English, translated into Dutch and French, and sent to all known British, French, and Dutch VCs in early 1992 (translations were retranslated into English to ensure correspondence of meaning). VCFs were identified using membership lists of national and European venture capital associations, trade journals, and Venture Economics' *Guide to European Venture Capital Sources*. We sent one survey to each of the VCFs identified in the three countries. Responses were 76 in the United Kingdom (43% response rate), 43 in France (25% response), and 37 in the Netherlands (40% response), 156 in all. Analyses of the characteristics of nonresponding VCFs show no significant differences from responding VCFs; thus, the probability of nonresponse bias appears minimal.

In the United States, personal interviews allowed us to encourage VCs to fill out surveys on two portfolio companies, preferably one doing well and one doing not so well. This procedure helped to limit possible sampling bias. In Europe, questionnaires were mailed to VCs who were asked to pick one venture only, but requirements that the VCs personally served on the boards and that the VCF's investment was at least two years old restricted their choice. The means of the performance rating in all countries was around the middle of the scale and their standard deviations were large enough to suggest that no systematic favoring of ventures by performance occurred. Table 2 presents descriptive statistics on the VCFs and portfolio companies in our sample.

Measures for Tests of Hypotheses

Dependent Variables

The central concepts in this study are the governance effort expended by VCs and the roles by which they add value to their portfolio companies. As has been noted by Rosenstein (1988) and Gorman and Sahlman (1989), among others, VCs influence and help direct venture efforts through personal involvement at board meetings and other informal interactions with CEOs. Accordingly, we have measured process governance (Sapienza 1994) as the frequency of face-to-face interaction in VC-CEO pairs: (1 = less than once a quarter; 2 = once a quarter; 3 = once a month; 4 = twice a month; 5 = once a week; 6 = twice a week; 7 = every day).

To measure value added we asked VCs to rate the importance (on a scale from 1 = not important at all, to 5 = of great importance) of the different roles they assumed with regard to the venture. These roles were derived from prior research (e.g., Gorman and Sahlman 1989; MacMillan et al. 1989) and our own field work. VCs also rated their effectiveness in carrying out these roles (from 1 = not at all effective, to 10 = extremely effective). By multiplying the importance by the effectiveness rating we derived a measure of value added for each role (with a possible range of 1 = no value added to 50 = very high value added). The overall value added then was the average score across the eight roles. Table 3 presents summary statistics on VC effort, VC-CEO interaction, individual value-added roles, and overall value added of VCs.

Independent Variables Predicting Governance

We used three indicators of agency risk, one of business risk, four of uncertainty, and two of VC experience to test our four hypotheses (H1–H4) regarding VC governance. As explained below, geographic distance was also used as a control variable in tests of both governance and value added. The means and standard deviations of these indicators appear in Table 4.

Agency Risk

Agency risk was measured by three variables: (1) the CEO's years of experience in the focal industry, (2) the CEO's previous start-up experience (coded "0" if the CEO had no prior start-up experience, "1" if he/she had), and (3) the number of years the venture had been in the VCF's portfolio. We expected that VCs would perceive agency risks as high when: (1) the CEO had relatively few years' experience in the venture's industry, (2) the CEO had relatively few years' experience in new firms, and (3) when the venture had been in the

TABLE 3 Means and Standard Deviations of Involvement Variables^a

Variable	U.K.		FR		NL		U.S.	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Measures of involvement								
Face-to-face interaction	3.43	1.00	2.67	0.99	3.06	1.13	3.20	1.05
VC's hours devoted to the venture per year	153.74	116.55	79.02	69.88	75.00	53.07	194.05	187.76
Value added in strategic roles								
Sounding board	35.74	9.89	28.59	11.11	34.00	8.15	36.89	10.23
	(4.4, 8.1)		(4.2, 6.8)		(4.3, 8.0)		(4.5, 8.1)	
Financier	33.45	12.95	35.72	10.96	33.94	13.62	32.58	15.50
	(4.2, 7.9)		(4.4, 8.2)		(4.3, 8.0)		(3.9, 7.7)	
Business advisor	31.50	10.38	27.97	9.78	26.71	11.78	34.40	9.20
	(4.2, 7.6)		(4.1, 6.9)		(3.6, 7.5)		(4.4, 7.8)	
Value added in interpersonal roles								
Mentor/coach	26.38	12.45	18.35	12.19	25.41	10.51	29.71	12.62
	(3.8, 7.0)		(3.4, 5.5)		(3.6, 7.0)		(3.9, 7.2)	
Friend/confidant	23.65	12.93	18.11	13.24	18.71	11.85	24.54	13.67
	(3.4, 6.9)		(3.0, 6.0)		(3.0, 6.3)		(3.3, 6.6)	
Value added in networking roles								
Source of professional contacts	17.50	12.77	19.21	11.37	21.12	10.23	20.28	11.67
	(2.9, 6.1)		(3.2, 6.0)		(3.1, 6.7)		(2.9, 6.1)	
Source of industry contacts	12.63	12.10	15.22	14.50	15.76	11.49	13.30	11.41
	(2.6, 4.9)		(3.0, 5.2)		(2.5, 6.3)		(2.6, 4.0)	
Management recruiter	17.49	14.71	10.50	12.33	18.94	14.10	17.47	11.45
	(2.8, 6.2)		(2.7, 3.9)		(3.1, 6.2)		(2.9, 4.8)	
Avg. value added	24.79	6.49	21.17	5.58	24.32	5.74	26.41	7.58
	(3.6, 6.8)		(3.5, 6.2)		(3.4, 7.2)		(3.6, 7.4)	

^a The mean importance and effectiveness ratings appear in parentheses below the mean value-added scores.

Operational definitions of the different roles presented in surveys:

Sounding board: Listens, responds objectively, frankly, and truthfully

Financier: Provides or arranges funding in a timely manner

Business advisor: Discusses plans, reviews targets, offers feedback, provides management assistance, notes threats

Mentor/coach: Provides encouragement, positive reinforcement, support, and motivation

Friend/confidant(e): Is concerned for [CEO], will go out of his way for [CEO], listens to [CEO's] problems

Source of professional contacts: Knows or can locate CPAs, lawyers, et al.

Source of industry contacts: Helps generate orders, reach licensing or lease agreements, locate key suppliers, etc.

Management recruiter: Helps locate key members for the management team

Abbreviations as in Table 1.

VCF's portfolio for relatively few years. Sapienza and Gupta (1994) also used CEO experience as a proxy for perceived agency risk. Using years in the portfolio as a measure of agency risk is consistent both with Larson's (1992) argument that social contracting can replace economic contracts over time and with Williamson's (1975) observation that over time repeated interaction limits opportunism. The three variables we chose represent agency risks emanating from the delegation of decision-making authority, rather than from goal conflict. Significant negative coefficients on these variables in our governance equations would support H1.

Consistent with Barney et al. (1989), we measured perceived business risk as VCs' satisfaction with the venture's recent performance. We used VCs' assessments of their satisfaction with venture performance (on a Likert scale from 1 = not at all satisfied to 5 = extremely satisfied) rather than sales or profitability measures because of the inadequacy of financial indicators for assessing the performance of venture capital-backed firms (Gladstone 1988). Given this rating scheme, greater business risk would be indicated by lower performance evaluations; thus, significant negative coefficients for venture performance would support

TABLE 4 Means and Standard Deviations of Independent Variables

Variable	U.K.		FR		NL		U.S.	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
CEO's industry experience	13.43	7.42	12.95	7.47	12.62	8.47	10.69	9.01
CEO's start-up experience	.36	.48	.44	.50	.33	.48	.50	.51
Venture's years in VCF portfolio	3.28	2.09	3.16	2.59	3.32	2.75	4.18	4.28
Venture performance	2.86	.95	3.23	.73	3.02	.75	2.96	.78
Venture stage	.74	.44	.86	.35	.89	.32	.55	.50
Technical innovation	.58	.50	.65	.48	.57	.50	.43	.50
Marketing innovation	.53	.50	.47	.50	.30	.46	.51	.51
Product innovation	.86	.35	.67	.47	.78	.42	.65	.48
Venture capitalist's VC experience	10.17	5.81	10.50	7.43	6.59	3.47	.38	.49
Venture capitalist's venture experience	2.31	1.24	2.79	1.17	1.86	1.15	2.14	1.22
Geographic distance	92.26	104.38	80.85	61.64	64.03	41.66	138.41	172.58

Legend: CEO industry experience = years of experience in the venture's industry; CEO start-up experience = 1 if CEO has had previous start-up experience, 0 if not; venture's years in VCF portfolio = years since VCF's initial investment in the venture; venture performance = averaged VCs' ratings of venture performance on 12 different financial and nonfinancial dimensions in Europe and 15 in the U.S.; venture stage = 0 if early stage, 1 if late stage; technical innovation = 0 if low relative to competitors, 1 if high; marketing innovation = 0 if low relative to competitors, 1 if high; product innovation = 0 if low relative to competitors, 1 if high; venture capitalist's VC experience = number of years in the venture capital industry (for U.S., 0 if <5 yrs., 1 if >4 yrs.); venture capitalist's new venture experience = number of years operating experience for the VC in the venture's industry (for U.S., the figure represents the number of rounds of financing for which the VC has worked with the venture in its industry); geographic distance = number of minutes it takes the venture capitalist to travel to the venture.

Abbreviations as in Table 1.

H2. Previous research comparing VCs' and CEOs' assessments of venture performance showed this measure to be highly reliable and valid (Sapienza 1992; Sapienza and Gupta, 1994).

Uncertainty

Uncertainty (defined as the gap between the information possessed by decision-makers and the information needed to make effective decisions [Galbraith 1973]) was operationalized by venture stage and by three measures of innovation. Sapienza and Gupta (1994) argued that uncertainty is at its height in the earliest stages of venture development. Because terminology and definitions of financing stage vary across and even within countries (Bygrave and Timmons 1992), we opted for a dichotimization of this variable (0 = early stage, 1 = late stage) to simplify interpretation and minimize the probability of observing spurious relationships. Our expectation that uncertainty is highest in early stage venture means that we were predicting a negative coefficient for the stage variable (H3).

Zaltman, Duncan, and Holbek (1973) suggested that uncertainty among decision-makers about cause-effect relationships increases as the level of innovation increases. Therefore, in order to capture uncertainty emanating from cause-effect uncertainty we used three measures of innovation consistent with the three types of innovation suggested by Zaltman et al. (1973): technical, product, and marketing innovation. VCs were asked to rate ventures' level of innovation relative to their competitors' on each of these three dimensions of innovation. For each, we coded responses as "0" if the innovation was rated equal to or less than competitors, "1" if rated greater or much greater than competitors. Because greater uncertainty here is indicated by higher numbers, positive values on these three indicators would support H3.

In Europe, we acquired two measures of VCs' experience: (1) VC experience was indi-

cated by VCs' years of experience in the venture capital industry, and (2) VCs' focal industry experience was indicated by their years' operating experience in the venture's industry. In the U.S. survey, the original measures had been slightly different: VC experience had been coded as "0" if VCs had worked in the venture capital industry for fewer than five years and "1" if it was five or more; VCs' focal industry experience had been indicated by the number of rounds of financing the VC had worked with the venture in the focal industry. Significant negative coefficients for these VC experience variables would support H4.

Independent Variables Predicting VC Value Added

All independent variables predicting face-to-face interaction were also used to predict value-added; the grouping of the variables varied slightly to reflect the constructs in the final three hypotheses. H5 predicts that VCs add most value when ventures are neediest; venture needs were assumed to be greatest when CEOs were inexperienced, when the venture had been in the portfolio a longer time, and when venture performance was perceived as being poor. H6 predicts that value added will be higher when uncertainty is greater. Early stage ventures face greater uncertainties regarding demand, pricing, product requirements, and channels of distribution, among others. Thus, H6 implies a negative relationship with venture stage but a positive relationship with the three types of innovation measured here. Finally, VCs' experience is predicted in H7 to be positively related to venture experience. As described in the prior sections, we measured experience as a venture capitalist and experience in the focal industry because the two represent two fundamentally different but potentially important ways in which VCs can add value.

Tests of Governance and Value-Added Hypotheses

The governance hypotheses and the expectations for value added were examined via ordinary least squares multiple regressions in which the 10 predictor variables and the control variable (i.e., geographic distance) were regressed against face-to-face interaction (see Table 5) and value added (see Table 6).

These tests were first conducted for each country separately as theoretical and empirical literature have indicated that significant differences exist between the United States and Europe and within the European venture capital market (Ooghe et al. 1991). Results within the United States appear in column 5 in both tables. Results within the United Kingdom, France, and the Netherlands appear in columns 1–3, respectively. We also ran tests with aggregated European data, using the United Kingdom as the base case and entering dummy variables for France and the Netherlands; these results appear in column 4. Aggregating the data in this way allowed us to see more directly how each compares with the other countries in Europe and to see how the overall models worked in the United States versus Europe. Pooling the U.S. and European data would not have been advisable given the differences in timeframe and terminology.

In conducting the regression analyses, we dropped all cases with missing data. Thus, missing data accounted for differences in the "Ns" across our tables. We included geographic distance (travel time for the VC to the venture) as a control variable in analyses because it may have had an impact on the cost of involvement (Lerner 1995) or the probability of opportunistic behavior (Barney et al. 1994b).

TABLE 5 Governance Tests (Predicting Face-to-Face Interaction)^a

	U.K.	FR	NL	EUR	U.S.
Adj. R ² , <i>n</i>	0.166, 68	0.127, 33	0.177, 30	0.179, 131	0.542, 33
Intercept	3.540***	4.114***	-0.241	3.303***	2.567**
Dummy Variables in All Europe Equation					
France				-0.693***	
Netherlands				-0.647**	
Agency Risks (H1)					
CEO's industry experience	-0.006	0.023	0.024*	-0.002	0.008
CEO's start-up experience	0.105	0.508+	0.646**	0.315*	-0.428
Venture's years in VCF portfolio	0.031	-0.114+	0.180	-0.015	-0.098+
Business Risk (H2)					
Venture performance	-0.121	-0.299	0.568*	-0.013	0.129
Uncertainty (H3)					
Venture stage	-0.649**	-0.411	-1.016+	-0.484*	-0.936**
Technical innovation	-0.077	-0.784*	0.521	-0.184	0.255
Marketing innovation	0.309	0.525+	0.483	0.187	-0.412+
Product innovation	-0.176	-0.632*	0.758**	0.009	-0.064
Venture Capitalist's Experience (H4)					
Venture capitalist's VC experience	-0.040+	0.001	-0.027	-0.020+	-0.646*
Venture capitalist's new venture experience	0.271**	0.066	0.050	0.174*	0.461*
Geographic distance	-0.002+	-0.056	0.000	-0.002*	-0.003*

^a Each column presents the unstandardized betas from an equation regressing the independent variables against face-to-face interaction in VC-CEO pairs in each of the four countries.

* $p < .10$; ** $p < .05$; *** $p < .01$, *** $p < .001$, one-tailed tests.

Abbreviations as in Table 1. EUR = Europe.

TABLE 6 Value-Added Tests^a

	U.K.	FR	NL	EUR	U.S.
Adj. R ² , <i>n</i>	0.209, 68	0.248, 33	0.145, 30	0.201, 131	0.345, 33
Intercept	11.076**	9.773**	22.700***	15.462***	9.087*
Dummy Variables in All Europe Equation					
France				-3.698***	
Netherlands				1.943+	
Venture Needs (H5)					
CEO's industry experience	0.027	0.138	-0.127	0.009	0.062
CEO's start-up experience	1.773	2.298	3.519	1.398+	-3.476
Years in the VCF's portfolio	0.316	-0.458	-1.248+	-0.141	0.087
Venture performance	1.819**	1.660	1.675	1.339**	3.013+
Uncertainty (H6)					
Venture stage	-1.385	3.626+	-7.149*	-0.911	-7.985*
Technical innovation	4.798**	2.228	-0.354	3.276***	1.413
Marketing innovation	2.345+	-1.768	0.154	1.384+	-0.583
Product innovation	0.220	0.064	3.272	0.544	-0.883
Venture Capitalist's Experience (H7)					
Venture capitalist's VC experience	0.267+	-0.098	-0.082	-0.146*	0.854
Venture capitalist's new venture experience	1.445*	2.439**	0.211	1.298**	3.320+
Geographic distance	0.004	-0.012	-0.146	-0.000	-0.021+

^a Each column presents the unstandardized betas from an equation regressing the independent variables against venture capitalist value-added in each of the four countries.

* $p < .10$; ** $p < .05$; *** $p < .01$, *** $p < .001$, one-tailed tests.

Abbreviations as in Table 1. EUR = Europe.

Additional Analyses

Tests for minimum tolerance limits indicate that multicollinearity does not pose a significant threat to interpretation of the regression results. As mentioned earlier, the matched pair sampling procedure used in the original study indicated a high level of interrater reliability between VCs and entrepreneurs. This correspondence reduces, but does not eliminate, concern about common method variance, which might hinder the interpretability of the data. Because in the present study we used only a single data source for responses, we used an analytic method suggested by Podsakoff and Organ (1986) to examine the extent to which common method variance might undermine the validity of our data. We examined the factor structure of the criterion and predictor variables. If only a single factor emerged or if the variance in the data were largely explained by a single factor, the threat of common method variance would be high. For our data, at least four factors emerged in each country with eigenvalues greater than 1, and the largest single factor in any country explained only 22% of the variance; the same pattern also held when all data were pooled. Thus, common method variance did not appear to be a significant problem for our data set.

RESULTS

Table 2 indicates the number and type of VCFs by country, the size of ventures' boards of directors (including the number of inside members), and several of the characteristics of the ventures themselves. Consistent with what has been observed in prior literature (e.g., Timmons and Sapienza 1992), our sample shows a much higher percentage of early stage ventures in the United States (43%) than in the United Kingdom (27%), France (14%), or the Netherlands (9%). Because economic sectors were measured differently in the two studies, we cannot make direct comparisons of ventures by economic sector between the United States and Europe. Manufacturing, computer-related, and consumer products account for 56% of the sample in the United Kingdom, 49% in France, and 35% in the Netherlands. Biotech, chemicals, and electronics make up 32% of the ventures in the Netherlands, but only 14% of those in the United Kingdom and France. Across the sample, the percentage of CEOs with prior start-up experience is between 30% and 40%. In all four countries ventures pursue greater relative product innovation than they do technical or marketing innovation.

Table 3 indicates that VC-CEO pairs averaged meeting slightly more than once a month. Our interviews showed that in the U.S. board meetings occur about once a month in very early stage ventures and quarterly as the venture reaches greater maturity. Thus, given that the majority of ventures in this sample are late stage ventures, the rate of face-to-face interaction in this sample probably indicates that VC-CEO pairs meet informally with some regularity. Table 3 also shows how many hours VCs devote yearly to each portfolio company. Clearly, VCs in the United States and the United Kingdom expend greater effort in monitoring and assisting their portfolio companies than do those in France and the Netherlands; this observation is true both in terms of the frequency of personal contact and the total number of hours expended on each venture.

Table 3 also displays a remarkable consistency across all four countries in regards to the value added through strategic, interpersonal, and networking roles. These roles are roughly equivalent to what Fredriksen et al. (1990, p. 251) labeled "competence, support, and access to networks." Generally speaking, VCs see themselves as providing the greatest value through their strategic roles and the least through their networking assistance. These results help to clarify how VCs add value to their portfolio companies. Much of the previous literature has failed to recognize the interpersonal dimension of VC-venture partnerships. Our

results indicate clearly that the social or interpersonal roles of venture capitalists are a significant element in the functioning of these partnerships. In the United States and the United Kingdom, value added through interpersonal roles is close to the level of value added in strategic roles, whereas in France and the Netherlands the value added of interpersonal roles is closer to the less important networking roles.

U.S. VCs see themselves as offering the greatest value added overall. The closeness of the average ratings between the United Kingdom and the Netherlands may be somewhat misleading in that VCs in the Netherlands place greater emphasis on networking roles, roles seen as least critical across the sample. In general, perceptions of value added correspond with effort expended (hours devoted to the portfolio company and frequency of interaction). Table 4 indicates that CEOs average between 11 and 13 years of experience. Venture performance ratings in all four countries are near the midpoint, indicating that the sample is not biased toward either high or low performing ventures.

Governance Results

Table 5 presents the results of regression equations testing our four governance hypotheses. As may be observed, face-to-face governance in the United States is much better explained by the factors examined in this study than is such governance in Europe. Specifically, the adjusted R^2 for the model in the United States is 54%; it ranges between 13% and 18% in the European countries, and 18% overall for Europe. We will first present the results comparing the U.S. and the pooled European sample; we will conclude with a summary of the impact of country on face-to-face interaction.

Agency Risk and Governance

H1 (face-to-face interaction is greater when agency risks are high) receives mixed support. Less CEO experience is not related to more face-to-face contact. CEOs' industry experience is unrelated to VC-CEO contact both in the United States and in the pooled European sample; in France, contrary to expectations, VCs more frequently visit CEOs with greater industry experience. Further, in the overall European sample (and in the Netherlands and France), CEOs with more start-up experience are visited by VCs significantly more frequently than less experienced CEOs. In the United States, the relationship is negative, as predicted, but is not significant. As predicted, fewer years in the VCF's portfolio are associated with significantly greater face-to-face contact in France and the United States. This relationship is relatively weak even in these countries and does not hold for the overall sample in Europe.

Business Risk and Governance

H2 (face-to-face interaction is greater when business risk is high) is not supported. Venture performance is unrelated to governance in the United States and in the pooled sample in Europe; contrary to our prediction, it is positively related to VC-CEO interaction in the Netherlands.

Uncertainty and Governance

H3 predicted that greater uncertainty leads to greater governance effort. This prediction is strongly supported in terms of venture stage but is not supported in terms of innovation. Earlier stage ventures receive significantly more face-to-face interaction both in the United

States and in Europe. It should be noted that this result holds despite controlling for the number of years the ventures have been in the portfolio. All three subsamples in Europe show the same strong negative relationship; in France, the relationship is also as expected but is not significant. The results do not indicate increased governance in response to innovation as predicted in the hypothesis. For the pooled European sample, none of the three measures of innovation are significantly related to VC-CEO contact. In the United States, only marketing innovation is related to face-to-face interaction, and it is in the opposite direction of the one predicted. Further, whereas marketing innovation in France and product innovation in the Netherlands are related to face-to-face interaction in the manner predicted, both technical and product innovation in France are in the opposite direction as that predicted.

VC Experience and Governance

As we predicted in H4, VCs' experience in the venture capital industry is negatively related to face-to-face interaction in both Europe and the United States. However, contrary to expectations, VCs' experience in the new venture's industry is positively associated with governance in both the United States and the pooled European sample. The strong relationships in the UK between VCs' experience in the venture's industry and personal contact appear to be driving the significant results in the pooled sample. Thus, H4 receives mixed support.

Distance and Governance

As might be expected, geographic distance appears to reduce face-to-face contact; this effect is significant in the United States, in the pooled European sample, and in the United Kingdom.

Governance by Country within Europe

The significant negative coefficients in column 4 of Table 5 for the country dummy variables (FR = -0.693 ; NL = -0.647) indicate that when compared with the United Kingdom and the Netherlands, VCs in France have significantly less face-to-face interaction with CEOs; and when compared with the United Kingdom and France, VCs in the Netherlands have significantly less interaction with CEOs. In other words, country has significant explanatory power for interaction in Europe. An inspection of Table 3 shows that what this means is that VCs in the UK interact on a face-to-face basis with their CEOs more frequently than do the VCs in the other two countries. As mentioned earlier, VCs in the United States and the United Kingdom appear to take a much more active governance role than do those in France and the Netherlands.

Value-Added Results

Table 6 presents the results of equations regressing value added against indicators of venture need, uncertainty, and VC experience. Table 6 shows that as with governance efforts, the value-added efforts of VCs in the United States is better explained than are those of European VCs. However, in this instance the difference is not as dramatic. Specifically, the adjusted R^2 for the model in the United States is 35%, whereas it ranges between 15% and 25% in the European countries and is 20% for the pooled sample in Europe.

Venture Need and Value Added

For the most part, H5 (that venture need is positively related to VC value added) is not supported. CEOs' industry experience is unrelated to value added in the United States and in Europe, including the subsamples in each country. As anticipated, CEOs' start-up experience is negatively related to value added in the United States, but the relationship is not significant. Start-up experience is marginally positively related to value added for the pooled sample in Europe, a result opposite that expected. Second, as predicted, fewer years in the portfolio are negatively related to value added in the Netherlands. However, this relationship is marginal, and years in the portfolio are not related to value added in the United States or the pooled European sample. Finally, venture performance is positively related to value added both in the pooled European sample and in the United States. This result is contrary to our expectations.

Uncertainty and Value Added

Our results indicate consistent support for the prediction that value added will be greatest when uncertainty is high (H6). As predicted, a strong negative relationship exists between venture stage and value added in the United States. No relationship exists between stage and value added for the pooled European sample, but this result is complex. The expected negative relationship between stage and value added is present in the Netherlands, but this is offset by an unexpected positive relationship in France. Interestingly, French VCs appear to interact more frequently with early stage ventures (see Table 5) but perceive greater contributions in later stage investments.

As anticipated, the results show a positive relationship between venture innovation and value added in the pooled European sample, but innovation is not related to value added in the United States. Value-added increases significantly with increases in technical innovation and marketing innovation in the pooled sample for Europe. These results appear to be largely due to strong relationships between innovation and value added in the United Kingdom. Product innovation has no impact on value-added in any of the four countries.

VC Experience and Value Added

H7, predicting that value added will be directly related to VC experience, receives mixed support. Contrary to expectations, VCs' venture capital industry experience is negatively related to value added in the pooled European sample; it is not related to perceptions of value added in the United States. As we predicted, however, VCs' experience in the venture's industry is positively associated with value added both in the United States and in the pooled European sample. Further, both types of VC experience are positively related to value added in the United Kingdom.

Distance and Value Added

Geographic distance is not related to value added in the pooled European sample or in any of its subsamples. Distance marginally dampens value added in the United States.

Value Added by Country within Europe

The significant negative coefficient in column 4 for France (FR) indicates that when compared with the United Kingdom and the Netherlands, VCs in France add significantly less value to their ventures. As was the case with face-to-face interaction, these results indicate that country holds significant explanatory power for value added in Europe. An inspection of Table 3 shows that VCs in the United Kingdom offer the greatest value of the three, closely followed by those in the Netherlands and, more distantly, by those in France.

DISCUSSION

This study provides insight into how and when VCs in the United States and Europe vary involvement in their portfolio companies and what the value-added consequences of this variation are. We extend prior work on governance in at least two directions: (1) we examined a broader conceptual framework of governance than previously used, and (2) we developed and tested a model of VC value added. Moreover, by collecting data in four major markets for venture capital, we were able to examine what elements of VC involvement are constant across settings and which ones vary. Thus, our results help to illustrate the nature and effect of variations of VC involvement on a country-by-country basis. Knowledge of differences and similarities in norms and practices rises in importance as competition among high-potential ventures becomes increasingly global.

Our results show that agency, business risk, and information-processing perspectives are useful in predicting VC monitoring behavior in all four markets. However, the weak results with regard to agency risks suggest that problems of delegation of decision-making authority and managerial opportunism may be adequately handled by the contractual and structural arrangements between VCs and their portfolio companies. As predicted, we found that VCs engaged in greater face-to-face oversight of their investments when ventures faced the uncertainties associated with early stage ventures, when the VCs themselves had less experience as VCs, and when the VCs were in greater geographic proximity to the venture. Contrary to expectations, VCs with greater experience in the venture's industry interacted more frequently with their CEOs, and, in Europe, VCs interacted more frequently with CEOs who had greater start-up experience. Poor venture performance did not appear to prompt greater monitoring as had been expected.

We developed a model of value added based on the expectation that VCs would add most value when ventures were relatively weak, when uncertainty was high, and when VCs were relatively more experienced. Results indicated that, as expected, VCs experienced in the focal industry provided significantly more value than those with less focal industry experience. However, contrary to expectations, VCs added more value to high performing than to poorly performing ventures. Additionally, experience in the venture capital industry was positively (but not significantly) related to value added in the United States but was negatively related to value added in Europe.

Multiple explanations appear feasible for results contrary to expectations. Our failure to observe the expected relationships between CEO experience and monitoring or value added may have to do with individual preferences or power. Barney et al. (1994a) found inexperienced CEOs more receptive to VC advice. It is possible, therefore, that VCs have to expend greater effort influencing experienced CEOs. A second possibility is that more experienced CEOs demand more attention, especially in Europe where such effort on the part of VCs appears lower. This possibility points up a potential weakness of the agency

perspective in its implicit assumption that the principal controls the nature and level of governance.

Perhaps the most fascinating results involve venture performance. We had argued that VCs would interact with and add more value to ventures doing poorly. Our results show that venture performance had no impact on face-to-face interaction, except in the Netherlands where the relationship is positive rather than negative. Further, venture performance was positively related to value added in both the pooled European sample and in the United States. Although business risk arguments suggest that VCs will be more involved when venture viability is threatened, an alternative view is suggested by prospect theory. In this view, decision-makers seek to avoid or limit risks when faced with a situation in which high returns are expected (Sitkin and Pablo 1992). Thus, it may be that when VCs begin to anticipate high returns from superior ventures, they do whatever they can to enhance and protect those returns—i.e., they pour more effort into adding value. This interpretation is consistent with Sapienza et al. (1994) who speculated that VCs might follow a “home run” strategy and focus attention on the winners in their portfolios rather than ventures that are expected to yield little return. Additionally, VCs may seek to add value to high performers because they prefer to be associated with a winner. On the other hand, we cannot rule out the possibility that when VCs are satisfied with venture performance they assume their input has been more worthwhile. Finally, we may simply have the direction of causality reversed. Perhaps when VCs add more value, the ventures do better.

Differences in our regression equations by country indicate that VCs were driven by different factors in different markets. Further, the significant coefficients for the country variables in the European analyses demonstrate that the level of monitoring and value added of VCs is at significantly different levels in the three countries. Clearly, VCs in France are least involved and add the least value to their portfolio companies; generally speaking, VCs in the United Kingdom are most involved and add the most value. Whereas comparisons between Europe and the United States must be made with caution for reasons explained in the Method section, indications are that VC behavior in the United Kingdom is more similar to that in the United States than are the other two. The numerous similarities across markets are captured by our theoretical perspectives and detailed above, but the substantial differences are consistent with Hofstede's (1984) view that culture plays a dramatic role in the conduct of business. Further, the fact that our models best predicted the behavior of VCs in the United States lends some support to Hofstede's contention that theories are culturally bound: the theories we used were largely developed in the United States.

A close inspection of the regression equations suggests that VCs in France behave least like those elsewhere. Not only are the French VCs least involved, but their monitoring pattern is hardly captured by the variables measured here, and they are the only country in which value added is perceived as being higher for later stage ventures. One explanation of these differences is that VCFs in France are dominated by financial institutions whose VCs may be primarily former bank managers with less experience in venture building. Another factor may be that institutional investing in France is provided by investors from outside France. Whereas we can only speculate about possible cultural effects on behavior, the fact that VCs in France were least like that of the others is consistent with Hampden-Turner and Trompenaars (1993, p. 15) who place the values of managers in the United States, the United Kingdom, and the Netherlands in one group and those in France, Japan, and Germany in another.

The most consistent result across the sample was that VCs saw their strategic roles as their most important contributions, followed by their interpersonal roles, and then their net-

working roles. This result is particularly notable, because so little attention has been given in the literature to the importance of the interpersonal aspects of the venture capital process (Fried and Hisrich 1995). The importance VCs accord to mentor and confidant roles provides additional insight into how VCs develop the desired receptivity in entrepreneurs (Barney et al. 1994a) and are thereby able to add value. One of the VCs we interviewed expressed the importance in this way:

In some cases, where the relationship is really a solid one with management, we can add a lot more [than just money]. In those cases where we don't have a good relationship—when we don't have rapport—there is just nothing we can do.

Anecdotal and empirical evidence (Sapienza 1989) suggest that by and large VCs are able to create and maintain the level of understanding and rapport necessary to function effectively. The evidence here suggests that those VCs who develop only distant or adversarial relationships may be seen as uncaring "vulture capitalists" and may not be allowed to perform their fiduciary and advisory duties effectively.

Implications for Practice

Our data support Bhidé's (1994) contention that the personal contact of outside investors with the management of private firms plays a role in the governance of those firms, and Fiet's (1995) view that VCs try to manage agency risks through face-to-face post-investment interaction. In our sample, the more developed the venture capital market, the more personally involved VCs were in their investment companies. Further, the more involved they were, the more they contributed to these firms. These observations hold implications for all VCFs, but the implications vary by country in some instances.

Our results suggest that the high contact level for VCFs in the United States may help to protect the U.S. market somewhat from entry by foreign VCFs. However, given the growing variety of sources of financing in the United States (Timmons and Sapienza 1992), such barriers may not be sufficient to stave off competition. VCFs in the United States may be increasingly pressed to seek other ways to differentiate themselves from domestic competitors and substitutes. The importance accorded VCs' interpersonal roles suggests that VCFs might build sustainable competitive advantage by selecting and developing new partners with skills and aptitudes especially suited to the strategic and human challenges faced by entrepreneurs. Our data also suggest that experienced VCs are more efficient in monitoring new ventures; however, value added appears to be most related to focal industry experience. Thus, VCFs should not expect new partners to be as efficient in these roles as they will eventually become. Of course, VCs must also learn to manage the balance between maintaining fiduciary objectivity and establishing a professional closeness. Further, it may be important for VCFs to specialize by industry or to bring in staff to supplement industry-specific knowledge.

Whereas these same implications exist for VCFs in the United Kingdom, some additional possibilities also exist. First, VCFs in the United Kingdom may be able to more fully penetrate the market for value-added money and assistance. As McNally (1994) points out, the contribution of the venture capital industry to the growth of new firms in the United Kingdom has been somewhat limited by its focus on MBO deals. By partnering with corporate investors who could add their technical expertise to the VCFs' strategic and development skills, VCFs could expand into more early stage, high technology investment opportunities. Further, the relative absence of high value-added activity by VCFs on the continent suggests that U.K. VCFs could successfully penetrate that market. Geographic distance will

be less an impediment for U.K. than for U.S. VCFs. Nonetheless, new entrants from either country would have to find ways of dealing with barriers posed by language and cultural differences.

As markets become more fully global, VCFs on the continent may be forced to contend with greater outside competition. Entrepreneurs in the Netherlands and France may be in a position to demand more than mere financial involvement for a stake in their ventures. Even if such competition from the outside does not materialize, VCFs within these countries may gain significant advantage from adopting the practices of U.S. and U.K. VCFs. A question we cannot answer from our data is whether the tremendous emphasis French VCs put on their financial role is consonant with the wishes of their entrepreneurs. Previous studies (e.g., Fredriksen et al. 1990; Landstrom 1991) suggest that entrepreneurs in parts of Europe are dissatisfied with the industry and operating experience of VCs and with their relatively low level of involvement in the venture building process. Whereas such competencies may routinely develop as these markets mature, those VCFs that pioneer this direction in investor activity may reap significant experience and reputational benefits.

Another implication of our findings is that entrepreneurs may be able to understand and anticipate the variation in their investors' oversight activities as well as the types of roles VCs see as most critical. Our data reveal that entrepreneurs can expect (1) more involvement when the venture is in an early stage of development; (2) more involvement when the venture is doing well; and (3) more value placed by VCs on their strategic and interpersonal roles than on their roles as contacts to outside networks. As venture capitalists do, entrepreneurs should put effort into selecting investment partners, and after selection, into developing productive exchange relationships with them.

The absence in this study of an association between value added and the CEO's experience is consistent with our earlier study in the United States (Sapienza 1992) and suggests that VCFs can add value regardless of the experience level of entrepreneurs. The assumption in the United States that experienced entrepreneurs cannot use assistance (Sapienza et al. 1994) is not shared by European venture capitalists who are more involved with experienced CEOs. The actual transfer of knowledge, however, may depend largely on entrepreneurs remaining receptive to advice and information exchange (Barney et al. 1994a).

This study also holds implications for policy-makers in all four countries. One implication is that if policy-makers are going to try to stimulate growth through venture capital, incentives should be aimed at stimulating more early stage venture capital where the greatest bang for the buck appears to be achieved. If entrepreneurial companies are truly the engines of economic growth and of the renewal of the industrial fiber, then policy-makers should direct special attention to providing incentives for VCFs' investment in early stage ventures. In the light of our findings, the recent decision by the Dutch government to discontinue its state guarantee scheme (the so-called PPM rule, in which a portion of losses incurred by VCFs can be recovered) appears to be counterproductive. As a direct consequence of this policy change, many PPM VC funds have ceased operations, diminishing the amount of capital available for start-ups and the supply of experienced VCs. Our results suggest that, instead, the guarantee scheme should have been broadened to also encourage investment in start-ups in other European countries (e.g., France) where VC involvement is minimal and early stage capital is unavailable.

Policy-makers might be well advised to provide incentives for managers with industrial experience to start or join VCFs and to assist in the training of recruits from the industrial sectors. Since much of the additional value of venture capital financing appears to come from the interpersonal relations themselves, training should include information not only about

the economic aspects of venture building but also about critical social and supportive roles. Finally, agencies could help entrepreneurs by establishing services that provide information on VCFs' investment preferences, performance, experience, and reputation. In countries where VCFs follow passive investment strategies, policy-makers might also consider legislation reducing impediments to cross-border venture capital investing.

Limitations and Future Research Directions

We urge the reader to view the comparisons of the United States and European markets with extreme caution. As we stated earlier, the separation in time of the two surveys may have caused unmeasured differences in results. Further, certain limitations exist with regard to all multiple country studies. Apart from the language difficulties, these studies have to acknowledge the differences in definitions and concepts that are used in different cultural settings. Such problems exist whether investigators are using primary or secondary data sources. Researchers would be well advised to seek to introduce variables to control for cultural differences, to include cross-country experts on their research teams, and to design into their instruments measures that counter existing differences in terminology or concepts. In essence, when conducting multiple country studies, knowledge of the different institutional and cultural environments is imperative to allow for a meaningful interpretation of the results.

This study was limited by its cross-sectional design and the self-report nature of the data. Although other or additional sources would have reduced concerns about common method variance, the nature of many of the variables precluded alternative data sources. Nonetheless, where possible, future research may profitably employ additional data sources and measures of some of the variables. Our study could not include all variables of possible consequence. For example, we did not measure the possible moderating effects of syndication. Another important variable might be VCF type. One might speculate that government-supported VCFs are hamstrung in their ability to attract high quality VCs, because they are not able to compensate VCs on a level equal to that of private VCFs. We were not able to test this possibility given the small number of government VC firms in our sample.

A more complete understanding of investor governance and involvement might be achieved by examining how VCs and entrepreneurs use structural and behavioral factors to define their relationships. Several studies have examined how VCFs adjust governance mechanisms to cope with increased agency, business, or uncertainty risks (e.g., Barney et al. 1989, 1994a, 1994b; Lerner 1995; Sapienza and Gupta 1994), but it remains a challenge of theoretical and practical importance to examine whether structural and behavioral mechanisms are used to complement or substitute for one another. In terms of the specific roles or activities undertaken by VCs, past studies have focused on the strategic and operational activities of VCs (e.g., Barney et al. 1994a; Harrison and Mason 1992; MacMillan et al. 1989). The results of this study suggest that deeper investigation into the importance and formation of interpersonal roles and their impact on monitoring effectiveness, entrepreneurial receptivity, and VC value added might also be productive.

Second, VC involvement behavior might also be more fully explained and predicted by approaching the issues from new perspectives. For example, one route might be to examine VCs as agents in addition to considering their positions as principals. VCs act as the agents of entrepreneurs in negotiating additional financing, selling the venture off to private interests, and taking ventures public. Further, VCs are agents of their limited partners. Bygrave and Shulman (1988) noted that VCFs receive annual fees based on the amount of capital

under management. This scheme provides them with an incentive to raise as large a fund as possible, regardless of whether they can adequately perform their fiduciary responsibilities over a greater number of ventures. This scheme also provides them an incentive to invest in later stage ventures since fewer such investments would be required to place all of their capital. As one VC said to us, "It doesn't take a genius to figure out that a 5% return on a \$100 million is better than 20% on \$10 million." Interviews also revealed that conflict of interest is often between one investor and another investor (rather than between investors and management as is assumed by agency theory) because of different preferences in the timing of exit or valuation of the venture. In short, both agency theory and understanding of investor-investee relations might be advanced by relaxing the unidirectional assumptions inherent in current applications of the theory.

Third, more research is needed on the impact of entrepreneur-investor dynamics on the effectiveness of the board of directors. Information theory and interactional justice theory might be used to help reveal how board meetings may be effectively managed, what information is most critical, and what interactional processes best transfer knowledge and capture the value-adding potential of outside board members. Further, research is needed to investigate the impact of entrepreneur-investor dynamics on venture strategy, including the level of aggressiveness, the level of innovation pursued, and the degree of risk-taking assumed by the venture.

Finally, the growing phenomenon of cross-border investing offers an attractive arena for study. How do geographic distance, venture stage, language, and cultural factors impact the nature and value of venture capital across borders? Our framework for value added suggests that a complementarity of background would enhance value added. Does a complementarity of cultural perspective also enhance value-adding potential? If a high-potential venture aims to be internationalized from the outset, would it benefit from a board of advisors/investors representing the various regions it is targeting? Although questions of governance and value added are essentially the same for cross-border relations as they are for single country ones, finding the answers may require additional methodological complexity and additional theoretical perspectives to the ones we have applied in the past.

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